

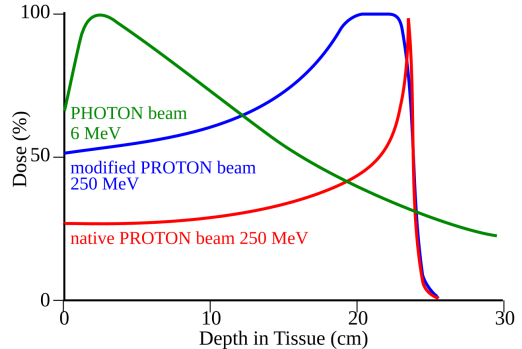
A High-Sensitivity In-Room CRYSP Scanner for Distal-Edge Verification in Proton Therapy

J.J. Gómez-Cadenas, F. López-Guejo and R. Soleti

BioGipuzkoa
September 9, 2026

What is proton therapy?

- A form of **external-beam radiotherapy** that uses accelerated protons to destroy tumour cells.
- Protons are **positively charged particles**: their energy determines how deep they travel in tissue.
- Their dose peaks near the end of their range (the **Bragg peak**), then falls sharply.
- Combining different proton energies covers the tumour volume, while **reducing dose beyond it** compared with X-rays.



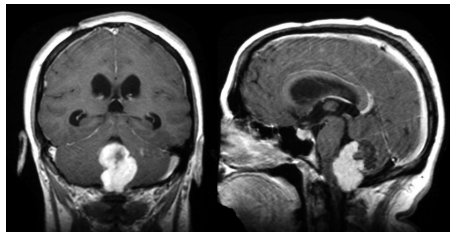
Schematic depth–dose profiles: a single proton energy, a spread-out peak and an X-ray beam.

Image: D. Ilyin / A. A. Miller, Wikimedia Commons (CC0).

Reference: National Cancer Institute, External Beam Radiation Therapy.

Example: Brain tumors

- Brain tumours are the **most common solid tumours in children**: about **300 diagnoses/year in Spain**, and also common in adults (~ 4500 cases per year).
- One example of brain tumour (ependymomas) often arise around the fourth ventricle, close to critical structures such as the **cerebellum and brainstem**.
- The goal and crucial challenge of radiotherapy is to **fully cover the tumour** with the prescribed dose while **sparing these nearby structures**.



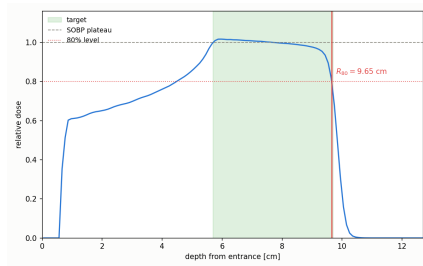
Fourth-ventricle ependymoma (MRI).

Sources: National Cancer Institute
Childhood Brain and Spinal Cord Tumors (PDQ);

Childhood Ependymoma.

Proton therapy: a sharp tool that must be aimed precisely

- The sharp dose fall-off allows full tumour coverage while sparing tissue beyond it — **in theory**.
- Unlike photons, protons stop inside the body. Their **distal falloff** must therefore be predicted using **analytical or Monte Carlo calculations**.
- The calculated range has a **systematic uncertainty of about 3.5%**. At 10 cm, adding 1 mm gives a **4.5 mm margin** (calculation on the right).
- A conservative range that spares nearby organs can **underdose a substantial fraction of a small paediatric tumour**.



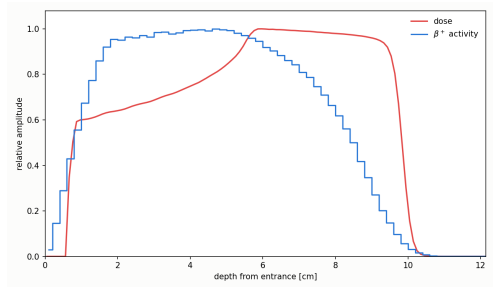
Spread-out Bragg peak (SOBP): flat dose over the tumour, sharp distal edge.

Margin at 10 cm range:

$$\begin{aligned} &0.035 \times 100 \text{ mm} + 1 \text{ mm} \\ &= 4.5 \text{ mm} \end{aligned}$$

PET: an *in vivo* probe of where the beam stopped

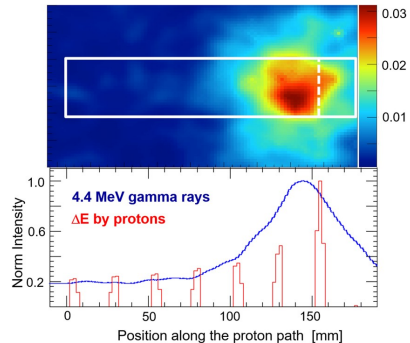
- Above the nuclear-reaction threshold, protons create β^+ **emitters** along their path — mainly ^{15}O and ^{11}C .
- Their annihilation γ 's are imaged in coincidence: a PET image of the **induced activity**.
- The activity falloff sits just upstream of the Bragg peak; its **distal edge tracks the range**.



Dose (Bragg peak) vs. β^+ activity and its distal edge.

Prompt gammas: tracking the proton range during irradiation

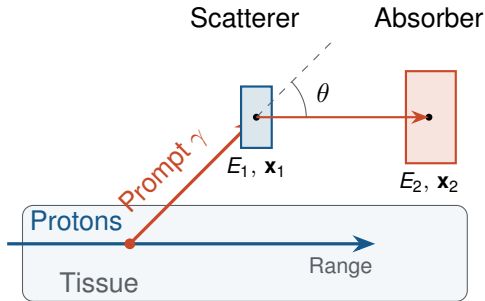
- Nuclear reactions along the proton path leave excited nuclei in the tissue. Their de-excitation emits **prompt γ rays** with energies of several MeV.
- Emission occurs **during irradiation**. Its distal falloff tracks the proton range; the relation depends on tissue composition and the selected γ energies.
- The measured **4.4 MeV γ profile** shown here concentrates near the Bragg peak and follows the proton range.



Measured prompt- γ profile and simulated proton energy deposition. 70 MeV protons; spaced PMMA slabs.

Imaging prompt gammas with a Compton camera

- A **Compton camera** measures the positions and energies of successive interactions of the **same γ ray**.
- The photon scatters in the first detector and is absorbed in the second. The deposited energies determine the **Compton scattering angle**.
- The two interaction positions define the cone axis; the scattering angle defines its opening. The emission point lies on this **cone of possible origins**.
- Combining many events reconstructs the **emission profile and its distal edge**.



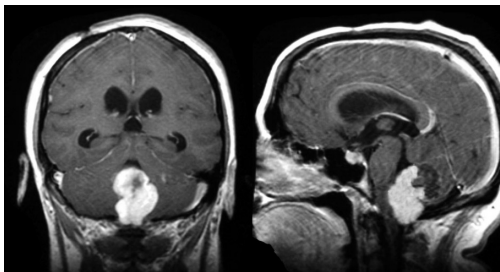
One photon, two detector interactions: Compton scattering followed by absorption.

Hueso-González et al., Front. Oncol. 6:80 (2016).

Hybrid in-beam PET and Compton imaging (2022).

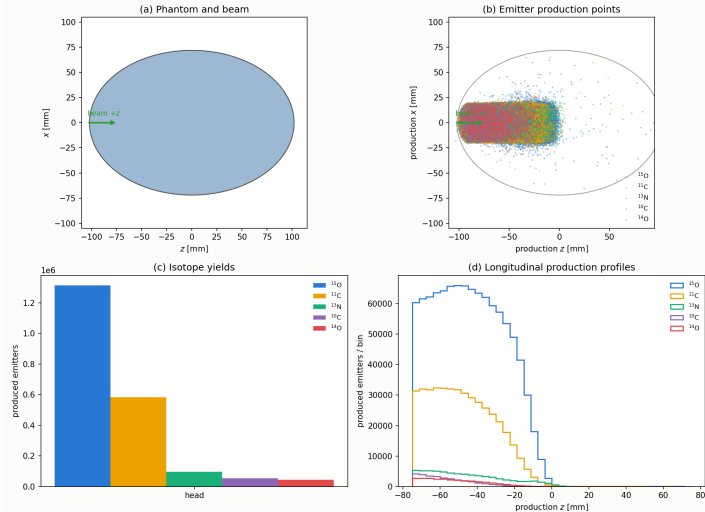
PET in proton therapy in brain

- Rigid, reproducible positioning — the target sits in the same place every fraction.
- Compact, well-defined anatomy: a short field of view covers it comfortably.
- High clinical stakes: paediatric and skull-base tumours next to critical structures.
- Our illustrative case: a fourth-ventricle ependymoma.

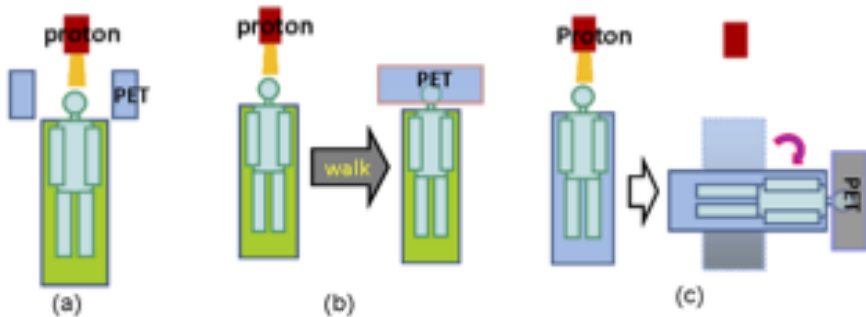


Fourth-ventricle ependymoma (MRI).

Head phantom used in the simulation



PET range verification: offline, in-beam and in-room



PET range verification: offline, in-beam and in-room

Offline

patient moved to
a nearby PET/CT

15–30 min later

washout, ^{15}O gone,
co-registration

In-beam

two planar heads
at the treatment
position

~ zero delay

limited angular
coverage

In-room

full ring, next
to the treatment
room

~ 2 min delay

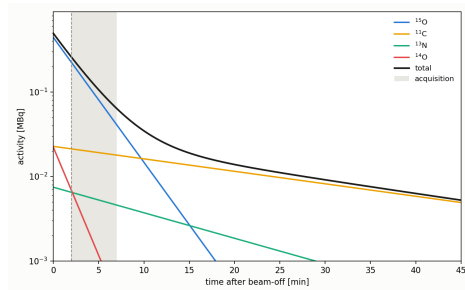
full coverage,
high sensitivity

The trade-off is **delay** vs. **angular coverage** / **scanner quality**.

Offline results in large uncertainties

A delay of 15–30 minutes costs on three fronts:

- **Sensitivity**: the abundant ^{15}O (2 min) has vanished — only the sparse ^{11}C tail remains.
- **Biological washout**: perfusion clears the emitters; hard to model, patient-dependent.
- **Co-registration**: the patient is set up afresh on a separate scanner — a few-mm misalignment maps straight into an apparent range shift.



^{15}O dominates the first minutes, then decays; ^{11}C carries the late, weak signal.

Online/In-room tradeoffs

In-beam (short delay, same bed)

- Retains the short-lived ^{15}O , essentially no delay.
- But the beam must pass: only **two planar heads** fit.
- Reduced angular coverage $\Rightarrow$ **low sensitivity** and limited-angle image artefacts.

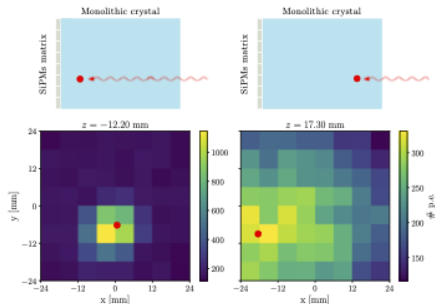
In-room (just after, full ring)

- A **full closed ring** $\Rightarrow$ full coverage, high sensitivity, clean image.
- Same couch $\Rightarrow$ a known rigid transform and a short transfer, without separate-scanner repositioning.
- The ~ 2 -min delay is **bought back** by sensitivity and a **better scanner**.

A **high-sensitivity in-room ring** combines short transfer time with complete angular coverage and high sensitivity.

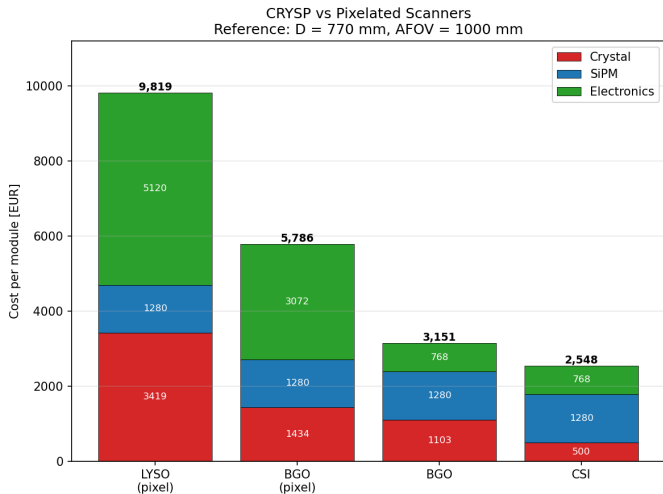
CRYSP: *CRYogenic Sensitive PET*

- Scintillators **CsI** and **BGO** gain dramatically in light yield when cooled — at 77 K, CsI reaches ~ 100 photons/keV and BGO ~ 29 photons/keV.
- In both cases, high yields translate into excellent energy resolution (6% CsI, 10% BGO), which in turn is essential for efficient scatter rejection.
- Monolithic crystals + SiPMs permit 3D position reconstruction using neural networks with high accuracy and no parallax error.

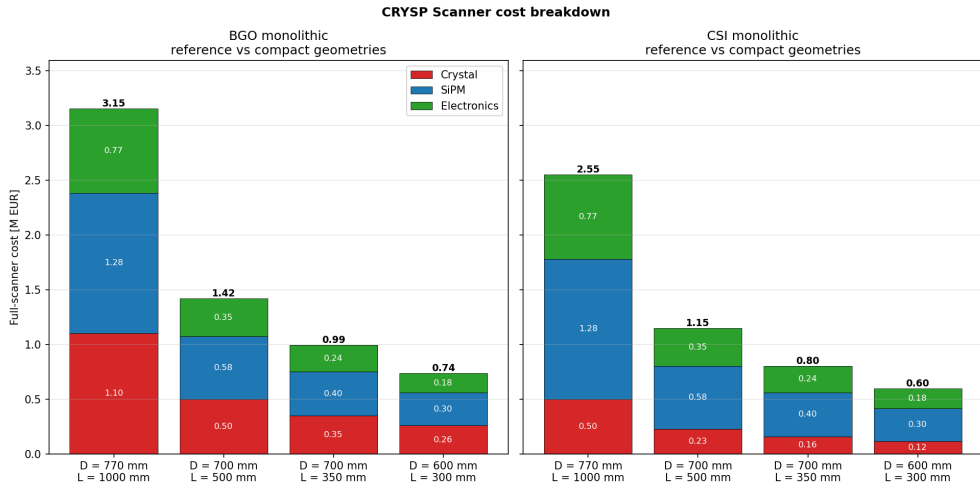


3D reconstruction in a monolithic crystal.

Scanner cost breakdown

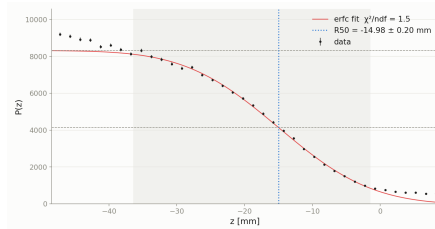
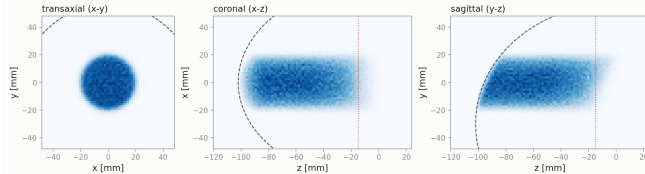


Total scanner cost



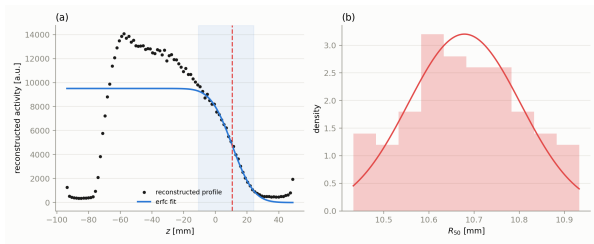
Imaging the brain

crisp_ring_1m_bgo_2x0 / bgo_195k_2X0 — run 000



Statistical procedure: from simulations to range precision

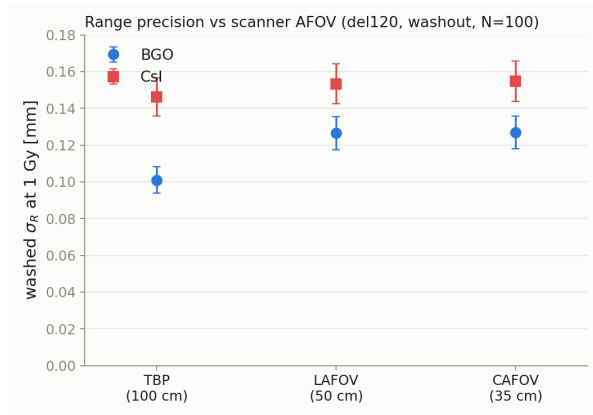
- Ten independent 1 Gy simulations provide the parent event pool.
- $N = 100$ independently thinned realisations estimate the run-to-run spread of the fitted activity midpoint, R_{50} .
- A finite-pool correction restores the count variance of independent acquisitions.
- Reference BGO: **0.070 mm nominal** and **0.101 mm with washout**.



Reference TBP BGO study: one shard and the washed
 $N = 100$ ensemble.

Reference protocol: all scanner configurations are sub-mm

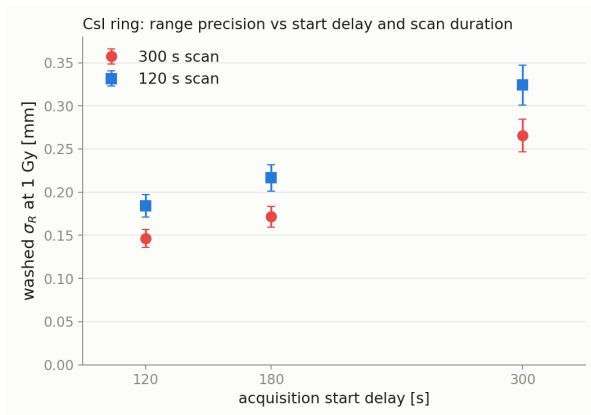
- **Protocol:** 1 Gy, 300 s acquisition, starting 120 s after beam-off; washout included.
- BGO: 0.101–0.127 mm; Csl: 0.146–0.155 mm.
- BGO is more precise through higher sensitivity; Csl remains a competitive, lower-cost option.
- These values are the **statistical precision of the fitted activity edge**.



Finite-pool-corrected σ_R for the reference protocol ($N = 100$).

Timing flexibility: the CsI stress case remains below 0.35 mm

- CsI is the lower-cost, less precise crystal; BGO is more precise at every configuration studied.
- At a 300 s start delay, σ_R is 0.266 mm for a 300 s scan and 0.324 mm for a 120 s scan.
- Across all tested delays and scan durations, $\sigma_R < 0.35$ mm.
- The scanner contribution is therefore well below the 1 mm target.



CsI range precision vs. start delay, with washout ($N = 100$).

Summary

- Proton-therapy range verification needs an **in vivo**, online/in-room, high-sensitivity probe
- **CRYSP in-room PET** exploits cryogenic CsI/BGO for excellent energy and spatial resolution at a **competitive cost**.
- For the reference protocol, the fitted activity edge has **0.101–0.155 mm statistical precision** across all BGO and CsI geometries; the tested CsI timing grid remains below 0.35 mm.
- CsI offers a very good, low-cost, supply-secure option; BGO offers the best statistical precision.

Thank you

Questions — and collaborations — welcome.